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Standalone fuzzy logic controller applied to greenhouse horticulture using Internet of Things

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***Abstract*— In this paper, a standalone Fuzzy Logic Controller (FLC) with Internet of Things (IoT) capabilities is developed for analysis and linguistic decision making about fertigation (fertilizers + water) in a greenhouse. The following input variables are evaluated: automatic drainage measuring, Electrical Conductivity (EC) and pH values of drainage. In order to determine if water content is easily available to crops, the lexical uncertainty given by "easily available" was quantified into a numerical value, as output is given a scalar value associated to fertigation. The result is a controller that embeds structural human knowledge about irrigation scheduling into an analytic FLC, in order to provide agricultural scientists with quick access to a particular crop's main production and growth related variables, and allowing for future data driven decisions on the spot. The controller was validated with real data collected from two particular crops, cucumber and greenpepper, using remote sensors connected to Internet. Final results seem support the continuing development of affordable yet efficient applications based on FLC and IoT, that provides effective production tools for horticulture producers in developing countries.**

Keywords*— Agriculture in protected environments, Fuzzy Logic Controller, greenhouse horticulture, Internet of Things.

I. INTRODUCTION

Without water, there is no life: this assertion applies not only to human life but to crops growing in protected environments. Since all human activities have been impacted by climate change effects, the Costa Rican government has comitted to having the country carbon neutral by 2021, as reported by [1]. A first approach for a Fuzzy Logic Controller (FLC) is presented in [2], which considers water as a critical input for greenhouse horticulture, agriculture and food production, being all severely affected by the excess or deficit of rainfall, changes in temperature and other associated environmental variations, which may be particularly extreme in fragile tropical ecosystems. These events may constitute serious threats to food security in low and medium income economies located in these regions, such as Costa Rica.

In [3], [4] a proposal is presented that uses Internet of Things (IoT) as a technological platform implemented with commercial off the shelf (COTS) components, to capture and analyze data for decision-making regarding climate change effects in greenhouse horticulture and new sources of energy as pellets out of a microalgae culture. A previous design, proposed by this paper's authors, implemented a FLC solution with two input variables, drainage and electric conductivity (EC), with fuzzy processing in the cloud, as shown in Figure 1 (see [5] for details). The current work extends the input variables to three, adding also an embedded FLC controller that takes care of the fuzzy operations, facilitating its work when no Internet access is available. The system is able now to take decisions on site while still serving as an IoT platform, showing graphical representations of its irrigation decisions via remote access devices.

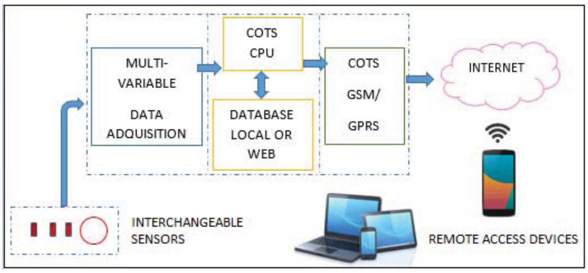

Fig. 1. Representation of the proposed general electronic embedded system with COTS components with IoT capabilities. Sensors are interchangeable according to the problem. A 32-bit ARM-based Lanin board (in the system's second version) is used as COTS CPU, with access to both a local and a remote web database. Data can may be visualized remotely thanks to the system's Internet connectivity and the use of IoT software such as ThinkSpeak. Data post-processing may be applied on the cloud, with results accessible from any computer or smartphone [3].

This paper is organized as follows: Section II describes the problems regarding irrigation decision-making considering linguistic terms related to water availabitilty, measuring drainage, EC and potential of hydrogen (pH). Section III presents the general structure of the electronic data collecting system, along with the application specific software layers needed. Some explanations are given on the IoT software platform being used and how it is integrated to the acquisition and processing system. In section IV the standalone FLC

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application in greenhouse horticulture is described. Finally, some conclusions and perspectives are given in Section V.

II. PROBLEM DESCRIPTION

One of the major problems faced by agricultural scientists and producers, when dealing with decisions aimed at improving crops yield or water usage, is the blurred boundaries between such lexical criteria as established in the technical literature. Besides, many of the percentages used to determine water or fertilizer quantities are based on rules of thumb, which sometimes are derived from years of experience on particular crops on particular environments. This makes difficult not only to compare or test the performance of any proposed method of irrigation or fertilization in heterogeneous environments, but also make cumbersome and error prone the tasks of monitoring and decision-making. And while the reliable measurement of variables related to crop growth and yield is not new nor expensive anymore thanks to the Internet [3],[5], not many abstract models are still available that may serve as quick decision tools for users.

A. Greenhouse horticulture

Nowadays, several recent studies have been carried out regarding wireless sensor networks applied to agriculture (see [6]) and greenhouses (see [7],[8]) which point out in the promising direction of exploiting IoT capabilities to carry out production decision-making on the cloud (see [9] for more on this). This has motivated the development of a general embedded system on an affordable platform, featuring Internet of Things (IoT) capabilities general enough for a wide span of potential applications. As an example of how such a system may help in the fight against climate change negative effects, two specific applications have been tested and validated by the authors: the first one oriented to horticulture production in protected environments [10]; the second application aimed at improving the biomass yield of a microalgae culture with potential uses in biofuel production [11].

III. A IOT DEVICE FOR MULTIDISCIPLINARY APPLICATIONS

The embedded system presented here were approached under a severe pair of restrictions: affordable for producers with limited economical means, and yet accurate enough for a researcher still exploring for alternative techniques of production. This requirements not only dictated the choosing of a COTS based system, but also promoted the use of online processing tools, that are commonly available at a low cost on the cloud. The result is a system under a thousand American dollars (with all sensors and connectivity included), but with a precision comparable to laboratory measurement equipment, and with data post processing available on the cloud. A first version of the processing board used a Arduino ATmega-based processor with bare metal coding. In the second version, code was migrated to a Lanín 32-bit ARM-based board [12] running under an embedded Linux Buildroot distro, with interchangeable sensors and data acquisition, wireless and 3G communication interfaces and at least two electromechanical actuators, and open source software running on the cloud (for data fusion and post processing enabling accurate monitoring and decision making). Using an embedded OS provided the system made it easier to configure depending on the final application (that usually means different sensors), via wired Ethernet, using a flexible HTML light web server running on the ARM board. The standalone FLC was tested in a Arduino one microcontroller in order to evaluate if the embedded Fuzzy Logic Library (eFLL) was capable to run efficiently in a small capacity controller and send the final linguistic decision to the IoT platform.

A. Connection

To allow for the required remote connectivity, a GSM/GPRS (Global system for mobile/General Packet Radio Service) was integrated to the system. The board --- an Arduino compatible shield adaptable also to the Lanin board--- is programmed to collect measured data and send it to an Internet service provider. Data is collected via the ThingSpeak platform [13]. ThingSpeak is an open source IoT applicacion and API (Application Programming Interface) that stores and retrieves data from things using HTTP protocol over the Internet.

IV. FUZZY LOGIC CONTROLLER

Figure 2 shows the complete system for acquiring data regarding crops production. The device includes a GSM communication board to connect to ThingSpeak platform. This is a standalone controller with IoT capabilites and automatic measurement of drainage, Ph, EC, ambient temperature and humidity.

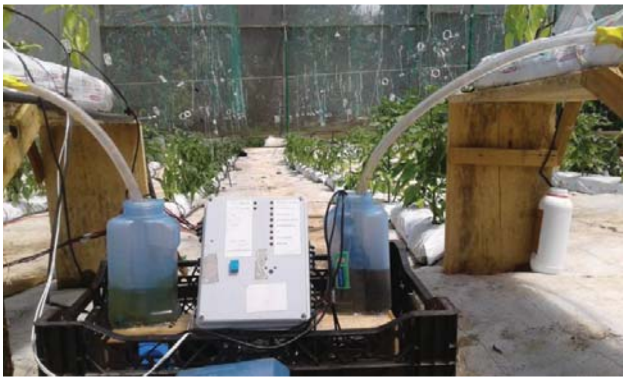

Fig. 2. Standalone controller with IoT capabilities, automatic measurement of drainage, pH, EC, ambient temperature and humidity. Collected data is send to ThingSpeak IoT platform [13].

A. Implementation of a Fuzzy Logic Controller

In this work, all the methodological steps to define a Mamdani type controller were followed. Figures 3 and 4 show the membership functions for input variables, with the fuzzy rules defined and the corresponding conditions. The controller output is a numerical value which corresponds to the linguistic variables related to the decision- making regarding irrigation.

To comply with step one of the methodology, three input variables are identified: drainage, EC and pH. For drainage triangular shapes were established to emphasize the points of measurement for decision-making, situation that is explicit in figure 3.

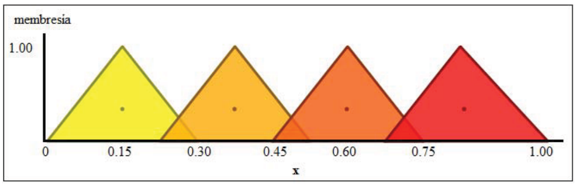

Fig. 3. Drainage membership function, triangular shapes were established to emphasize the points of measurement (from left to right): water very difficult available, water difficult available, water easily available and water very easily available.

EC and pH memberships were defined considering trapezoidal shapes. Figure 4 presents the selected ranges for EC measurements. All inputs, drainage, EC and pH are defined as "x" in the membership figures. For pH membership function five trapezoidal ranges were given: very acid, slightly acid, ideal pH, slightly alkaline and very alkaline, see figure 6.

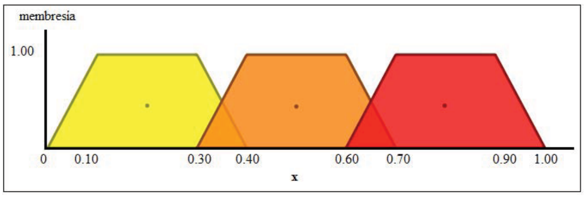

Fig. 4. EC membership function, three trapezoidal shapes were established to emphasize the given ranges (from left to right): low EC, good EC and high EC.

To comply with step two of the methodology, one output variable was defined as "Irrigation", triangular shapes were established for this membership (see Figure 5). This output was identified with the letter "y".

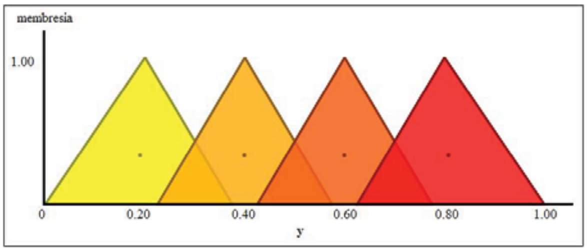

Fig. 5. Connection Irrigation membership function, triangular shapes were established for this output (from left to right): no irrigation, low irrigation, normal irrigation and high irrigation.

The output of the FLC gives a scalar value that is considered as a suggested value for irrigation decision-making. For example for an output value of 0.367 in low irrigation and 0.133 in normal irrigation, the final irrigation decision should be 13.3% of normal irrigation value, because water is easily available, CE is good and pH is ideal. With a numerical value as output the lexical uncertainty is easily solved with the FLC. See figure 6 with FLC results using QtFuzzyLite software version 6.0, the FuzzyLite Libraries for Fuzzy Logic Control [14].

B. Embedded Fuzzy Logic Library

In order to have a single hardware platform that allows decision-making, the embedded Fuzzy Logic Library (eFLL) [15] was selected to be included into the embedded system. It should be noted that other options were evaluated, for example the use of the Fuzzy Logic Library provided by MATLAB that can be used inside the ThingSpeak platform, but this alternative had an important limitation which is the dependency in a proprietary software and costs associated with the purchase of licenses for its use and update.

There is big pressure worldwide on water consumption [16], researchers and producers must come up with new efficient and scientifically tested alternative food production practices under the prevailing changing environmental conditions of our planet, including new guidelines for water saving, management and proved algorithms to take decisions at the right time. The use of IoT capabilities, thus, should provide with an efficient way of enhancing irrigation decision-making in order to save water and nutrients given to crops. One must consider, though, that vegetable production in greenhouses represents a challenge for typical farmers because there the growing behavior of most vegetable species is markedly different than in open air, and usual culturing practices must be modified accordingly [17].

C. Automatic data acquisition and filtering

Typically all measurements are done by an assistant once a day, very early in the morning. But in tropical countries, even under protected environments such as greenhouses, environmental conditions suffer rapid variations that, even though not extreme, affect severely certain crops [18].

In this case, a device was developed in order to automatize drainage collection procedures, excellent results were obtained using Kalman filter in order to be able to include data with high precision in the FLC (see [10], [19], [20], [21], [22] for details and considerations taken). Figure 6 shows an example of the operation of the whole Fuzzy Controller using QtFuzzyLite 6.0 [14].

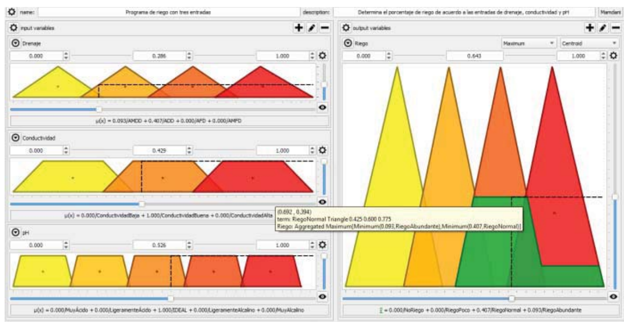

Fig. 6. Irrigation decision considering drainage (water difficult available), EC (good EC) and pH (ideal pH), output (normal irrigation). QtFuzzyLite 6.0 FLC was configured with membership function and FL rules [14].

D. eFLL programming structure

The eFFL library is written in C ++/C, and only requires the standard C language library "stdlib.h", so eFLL is a library designed not only for Arduino [23], but any embedded system that is written with C commands.

Listing 1 shows part of initialization code where the fuzzy objects and fuzzy sets are defined for both inputs and outputs. The code also shows trapezoidal and triangular settings, as well as the given names to each fuzzy set.

```
#include <FuzzyRule.h>
#include <FuzzyComposition.h>
#include <Fuzzy.h>
#include <FuzzyRuleConsequent.h>
#include <FuzzyOutput.h>
#include <FuzzyInput.h>
#include <FuzzyIO.h>
#include <FuzzySet.h>
#include <FuzzyRuleAntecedent.h>
Fuzzy* fuzzy = new Fuzzy();
//pH
FuzzySet* muy_acido = new FuzzySet(0, 0.025, 0.175, 0.2);
FuzzySet* ligeramente_acido = new FuzzySet(0.190, 0.215, 0.365, 0.390);
FuzzySet* ideal = new FuzzySet(0.380, 0.405, 0.555, 0.580);
FuzzySet* ligeramente_alcalino = new FuzzySet(0.570, 0.595, 0.745, 0.770);
FuzzySet* muy_alcalino = new FuzzySet(0.760, 0.785, 0.935, 0.960);
//conductividad
FuzzySet* baja = new FuzzySet(0, 0.1, 0.3, 0.4);
FuzzySet* buena = new FuzzySet(0.3, 0.4, 0.6, 0.7);
FuzzySet* alta = new FuzzySet(0.6, 0.7, 0.9, 1.00);
//drenaje
FuzzySet* agua_muy_dificilmente_disponible = new FuzzySet(0, 0.15, 0.15, 0.3);
FuzzySet*agua_dificilmente_disponible = newFuzzySet(0.225,0.375,0.375, 0.525);
FuzzySet* agua_facilmente_disponible = new FuzzySet(0.450, 0.6, 0.6, 0.750);
FuzzySet*agua_muy_facilmente_disponible=newFuzzySet(0.675,0.825,0.825, 1.00);
 // FuzzyOutput
FuzzyOutput* riego = new FuzzyOutput(1);
FuzzySet* no_riego = new FuzzySet(0, 0.2, 0.2, 0.375);
riego->addFuzzySet(no_riego);
FuzzySet* riego_poco = new FuzzySet(0.225, 0.4, 0.4, 0.575);
riego->addFuzzySet(riego_poco);
FuzzySet* riego_normal = new FuzzySet(0.425, 0.6, 0.6, 0.775);
 riego->addFuzzySet(riego_normal);
FuzzySet* riego_abundante = new FuzzySet(0.625, 0.80, 0.80, 1);
 riego->addFuzzySet(riego_abundante);
fuzzy->addFuzzyOutput(riego);
```

Listing 1. Code to call eFFL objects and initialize variables.

As mentioned in section III, the eFLL was programmed and tested inside a Arduino One microcontroller, results were compared against QtFuzzyLite output, with decision-making consistent in both systems.

In figure 7 are shown four different decisions given by the standalone FLC: output 2 has a 0.19 value which represents no irrigation, 0.50 at output 3 is for normal irrigation, output 4 with 0.32 means low irrigation a 0.81 scalar value in output 12 is for a high irrigation decision, as represented in the values over the central line for irrigation decision.

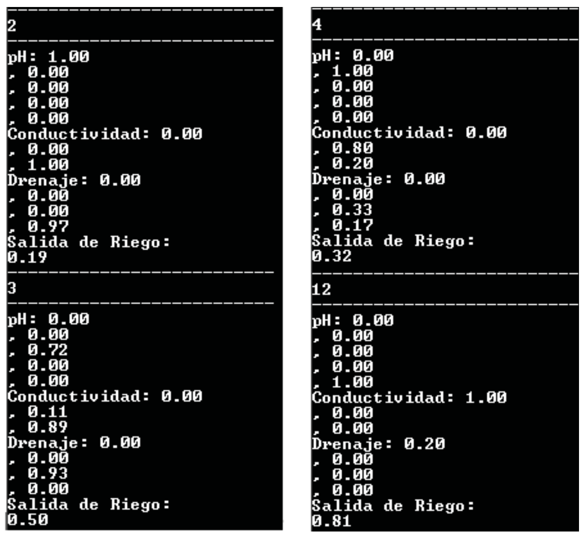

Fig. 7. Examples of different irrigation decisions taken by the standalone FLC, output given by an Arduino one microcontroller [23]

Figure 8 shows the final irrigation decision in the ThingSpeak platform, with data sent by the standalone FLC controller. These results are applicable to microalgae cultivation process, as the ones mentioned in [24], [25], [26], [27], [28], [30]. In [31], [32] some examples of crop water requirements and irrigation scheduling with conventional process are mentioned, where the proposed standalone FLC is also applicable to replace conventional techniques for decision-making [33].

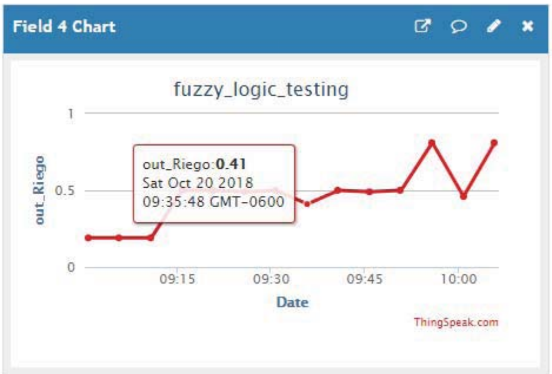

Fig. 8. Irrigation decisions given by ThingSpeak platform, output value 0,41 represents the scalar number for decision-making [33].

V. CONCLUSIONS

A complete electronic embedded system, used to automatically enhance overdrain measurement accuracy in greenhouse horticulture, with IoT applications capable of storing and analyzing multiple measurement of environmental variables related to crop production in greenhouses, and including a standalone FLC for decision-making regarding irrigation has been presented.

A Fuzzy Logic Controller (FLC) for decision-making about irrigation scheduling was programmed using eFLL and shown running in a standalone device, using an Arduino One microcontroller, that sends data to ThingSpeak IoT platform. The system was tested in a real greenhouse with vegetables crops. Researchers at Agricultural Engineering Department at Tecnológico de Costa Rica are collaborating with the post-processing data analysis in order to evaluate the algorithm's performance. The FLC output has been compared against QtFuzzyLite 6.0 and results are consistent with expert criteria in both controllers.

The developed system can be used for many other applications, such as: microalgae culture, forests and open air plantation monitoring, and even electrical smart grid monitoring. Only the sensors need be replaced for the intended application, and the data processing capabilities of the cloud can be used even for more specific applications, such as remote control and intelligent decision platforms.

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